Design and Verification of 32-bit ALU

Introduction

This project implements a 32-bit Arithmetic Logic Unit (ALU) using Verilog HDL. The ALU supports arithmetic and logical operations such as addition, subtraction, AND, OR, XOR, NOT, shift operations, and comparison.

Tools Used

- Verilog HDL
- Online Verilog Simulator

Operations Supported

- Addition
- Subtraction
- AND
- OR
- XOR
- NOT
- Shift Left
- Shift Right
- Comparison

Functional Verification

The ALU was verified using a Verilog testbench and simulation outputs were analyzed successfully.

Learning Outcomes

- RTL Design
- Verilog Coding
- Functional Verification
- Digital Design Concepts